25901036 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Minor Pathway Courses (MPC) - MG1MPCCHY100 - BASIC THEORETICAL AND ANALYTICAL CHEMISTRY

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

ONE SENTENCE

Answer any **10** questions

Each question carries **1** marks

1. Write the full ground-state electronic configuration for the element Sulphur which has an atomic number of 16. [K] / [CO1]
2. Provide an example of an element that shows extra stability due to a completely filled configuration, and write its actual electron configuration. [K] / [CO1]
3. Write down the de Broglie equation and explain the terms. [K] / [CO1]
4. Which of the three rules prevents a third electron from entering the 1s orbital of a Helium (He) atom? [K] / [CO1]
5. What is meant by the fine spectrum of Hydrogen? [K] / [CO1]
6. What is the function of the principal quantum number 'n' in defining the allowed orbits and energy levels based on the postulates? [K] / [CO1]
7. Describe the shape of an s-orbital [K] / [CO1]
8. Draw the formation of nitrogen molecule using Lewis dot structure. [K] / [CO2]
9. How electronegativity vary across a period? [K] / [CO3]

10. Define Atomic Radius. [K] / [CO3]

11. Give an example for an acid - base titration. [U] / [CO4]

12. Write the Lowry - Bronsted concept of acids and bases. [K] / [CO4]

[1x10 = 10]

Part B

SHORT ANSWER

Answer any **4** questions

Each question carries **5** marks

13. Write notes on (a) magnetic quantum number (b) spin quantum number. [K] / [CO1]

14. Explain briefly principal quantum number. [K] / [CO1]

15. What are the factors favouring the formation of ionic bond? [K] / [CO2]

16. Calculate the molality of a solution prepared by dissolving 0.150 moles of sodium chloride (NaCl) in 0.75 kg of a solvent. [U] / [CO4]

17. Write notes on (a) simple distillation (b) fractional distillation. [K] / [CO4]

18. Describe with examples primary standard and secondary standard. [K] / [CO4]

[5x4 = 20]

Part C

ESSAY

Answer any **2** questions

Each question carries **10** marks

19. Give notes on (a) orbit and orbital (b) Heisenberg's uncertainty principle. [K] / [CO1]

20. Write notes on (a) Hydrogen bond (b) Van der Waal's forces. [K] / [CO2]

21. Write an essay on Ionisation enthalpy. [K] / [CO3]

22. Give notes on (a) Normality (b) Mole fraction (c) ppm and (d) ppb. [K] / [CO4]

[10x2 = 20]